

STA 332/MATH 343: Midterm – Solutions

Duke University, Spring 2026

Instructions: You have 75 minutes to complete the following questions. This exam is closed book and no electronic devices are allowed. You must justify all assumptions, clearly cite any theorems used, and *show all your work*.

Your name: _____

(Turn over when exam starts)

Problem 1 [5+5+5=15 points]

Provide a brief (but formal) definition of the following concepts:

(a) Minimal sufficient statistic.

A statistic $T(\underline{X})$ is minimal sufficient if it is sufficient and for any other sufficient statistic $S(\underline{X})$, there exists a function h such that $T = h(S)$.

(b) Complete statistic.

A statistic T is complete for $\{f_\theta\}$ if for every function g , $E_\theta[g(T)] = 0$ for all θ implies $g(T) = 0$ with probability 1 for all θ .

(c) UMVUE.

An estimator $\delta(\underline{X})$ of $g(\theta)$ is a UMVUE if it is unbiased and has variance no larger than any other unbiased estimator for every θ .

Problem 2 [15 points]

Let $T = (T_1, T_2)$ be a sufficient statistic for θ in a model $\{f_\theta\}$. Which of the following statistics are we *sure* to be sufficient for θ ? (Briefly justify each answer.)

$$S_1 = (T_1 + T_2, T_1 - T_2), \quad S_2 = (T_1^2, T_2), \quad S_3 = (T_1, T_2 \cdot \theta), \quad S_4 = (T_1, e^{2T_1 + T_2})$$

We use the fact that if T is a sufficient statistic and $T = g(T^*)$ for some function g , then T^* is a sufficient statistic.

- S_1 is sufficient since $(t_1, t_2) = \left(\frac{s_1 + s_2}{2}, \frac{s_1 - s_2}{2}\right)$.
- S_2 is not necessarily sufficient since we may not be able to recover T_1 from T_1^2 (we don't know the sign).
- S_3 is not a statistic since it depends on θ .
- S_4 is sufficient since $(t_1, t_2) = (s_1, \log(s_2) - 2s_1)$.

Problem 3 [6+6+8=20 points]

Let $X_i \stackrel{\text{iid}}{\sim} f_\theta(x)$ for $i = 1, \dots, n$, where

$$f_\theta(x) = \frac{1}{\theta} \mathbf{1}_{\{\theta < x < 2\theta\}}, \quad \theta > 0.$$

(a) Prove that θ is a *scale* parameter.

The density of X can be written as

$$f_\theta(x) = \frac{1}{\theta} \mathbf{1}_{\{\theta < x < 2\theta\}} = \frac{1}{\theta} \mathbf{1}_{\{1 < x/\theta < 2\}} = \frac{1}{\theta} f(x/\theta),$$

where $f(x) = \mathbf{1}_{\{1 < x < 2\}}$. Thus θ is a scale parameter.

(b) Find a minimal sufficient statistic for θ . Justify your answer.

The joint density is

$$f_\theta(\underline{x}) = \prod_{i=1}^n \frac{1}{\theta} \mathbf{1}_{\{\theta < x_i < 2\theta\}} = \theta^{-n} \mathbf{1}_{\{\theta < x_{(1)} < x_{(n)} < 2\theta\}}.$$

By the factorization theorem, $T = (X_{(1)}, X_{(n)})$ is sufficient. We can see that if $T(\underline{x}) = T(\underline{y})$, then $f_\theta(\underline{x})/f_\theta(\underline{y})$ is constant in θ . Conversely, if $T(\underline{x}) \neq T(\underline{y})$, then $f_\theta(\underline{x})/f_\theta(\underline{y})$ is not constant in θ . Thus T is minimal sufficient.

(c) Is it complete? Justify your answer.

No. Since θ is a scale parameter, the distribution of $X_{(n)}/X_{(1)}$ does not depend on θ , so $X_{(n)}/X_{(1)}$ is ancillary. Let $c = E[X_{(n)}/X_{(1)}]$, which is a constant not depending on θ . Let $g(T) = X_{(n)}/X_{(1)} - c$. Then $E[g(T)] = 0$ for all θ , but $g(T)$ is not zero with probability 1.

Problem 4 [50 points + 10 additional points]

Let $X_i \stackrel{\text{iid}}{\sim} f_\theta(x)$ for $i = 1, \dots, n$, where

$$f_\theta(x) = \theta x^{\theta-1}, \quad 0 < x < 1, \theta > 0.$$

- (a) Determine the distribution of $Y = -\log(X)$ for $X \sim f_\theta(x)$.

Since $-\log(x)$ is a monotone function, by the change of variable formula, the density of Y is

$$f_Y(y) = f_X(e^{-y})e^{-y} = \theta e^{-\theta y}, \quad y \geq 0.$$

Thus $Y \sim \text{Exp}(\theta)$.

- (b) Find a sufficient and complete statistic T for θ .

Since this is an exponential family distribution with $R(x) = -\log(x)$, we have that $T = \sum_{i=1}^n -\log(X_i)$ is sufficient and complete for θ .

(c) Compute $E[T]$.

Since $Y_i \sim \text{Exp}(\theta)$, we have $E[-\log X_i] = E[Y_i] = 1/\theta$, so $E[T] = n/\theta$.

(d) Find the UMVUE for $1/\theta$.

We have $E[T] = n/\theta$, so $E[T/n] = 1/\theta$. Since T/n is unbiased and a function of the complete sufficient statistic, it is the UMVUE for $1/\theta$.

- (e) Find the MLE for $1/\theta$.

The log-likelihood is

$$\ell(\theta) = n \log \theta + (\theta - 1) \sum_{i=1}^n \log(X_i).$$

Differentiating and setting to zero gives

$$\frac{n}{\theta} + \sum_{i=1}^n \log(X_i) = 0 \implies \hat{\theta} = -\frac{n}{\sum_{i=1}^n \log(X_i)} = \frac{n}{T}.$$

So the MLE for θ is n/T , and the MLE for $1/\theta$ is T/n .

- (f) (Bonus 10 points) Determine the distribution of T .

Note that exponential distribution is a special case of gamma distribution, i.e. $Y_i \sim \text{Gamma}(1, \theta)$, so $T = \sum_{i=1}^n Y_i \sim \text{Gamma}(n, \theta)$.

